Network Scanning and Traffic Analysis Lab

Final Project Report

Prepared by: Rashed Rahman

Executive Summary

This project demonstrates both network reconnaissance and packet-level traffic analysis in a controlled virtual lab environment. Nmap was used to identify exposed services and attack surfaces, while Wireshark was used to observe how scanning activity appears at the packet level. The results highlight how attackers probe systems and how defenders can detect such behavior.

1. Introduction

Cybersecurity operations require both active scanning and passive monitoring. This lab integrates these two approaches to provide a complete understanding of network behavior from both offensive and defensive perspectives.

2. Objectives

• Identify live hosts
• Discover open ports and services
• Perform OS detection
• Capture and analyze network traffic
• Understand scanning behavior at the packet level

3. Lab Environment

Kali Linux (192.168.56.101)
Metasploitable 2 (192.168.56.102)
VirtualBox Host-only Network

4. Tools Used

Nmap – scanning and enumeration
Wireshark – packet analysis
Linux tools – ping, ip a
VirtualBox – lab environment

5. Methodology

5.1 Network Scanning

Host discovery, port scanning, service enumeration, and OS detection were performed using Nmap. Screenshots added to demonstrate results.

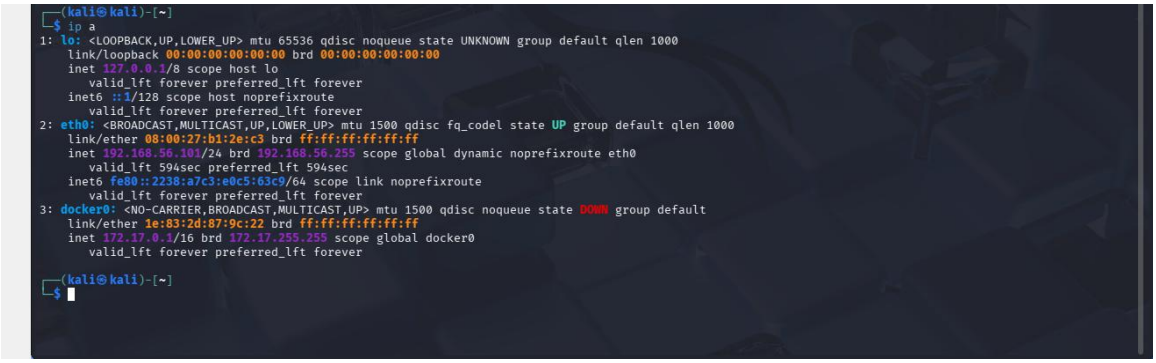

Figure 1: Kali Linux IP Address Configuration (ip a)

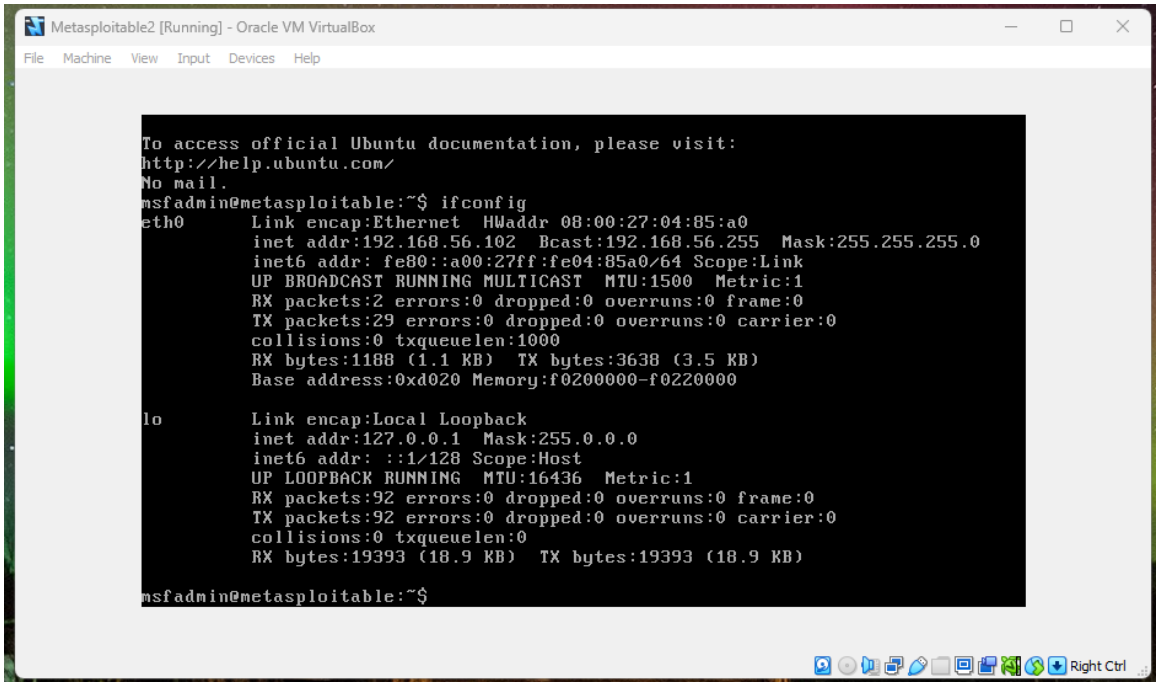

Figure 2: Metasploitable 2 IP Address (ifconfig)

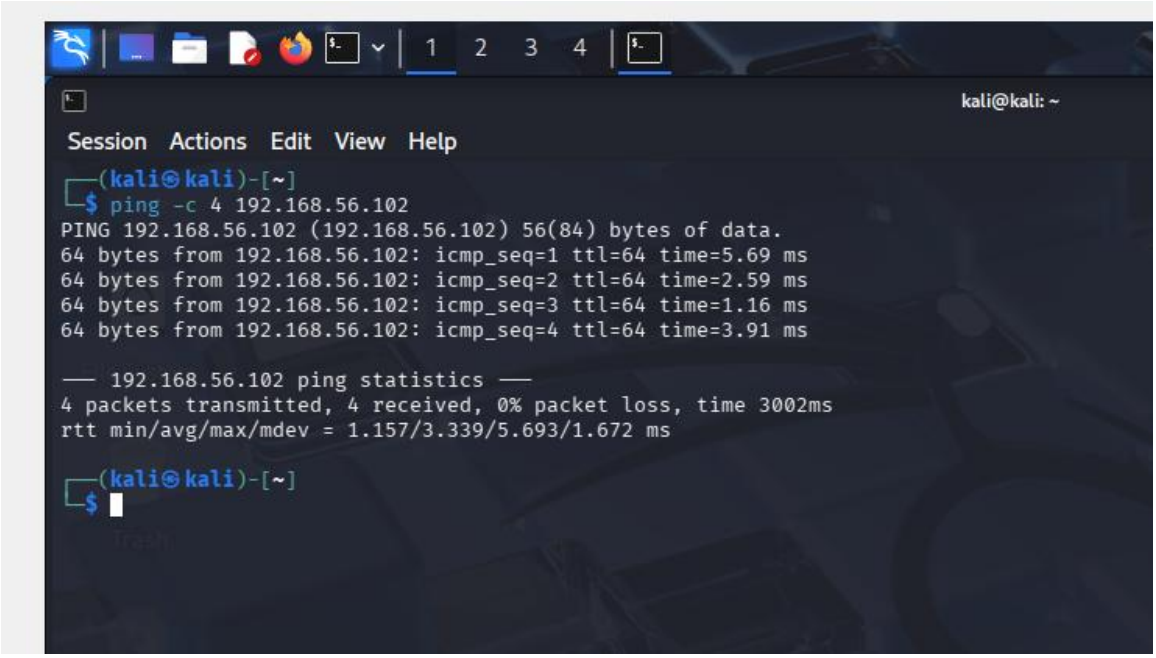

Figure 3: Successful ICMP Ping Test Between Kali and Metasploitable

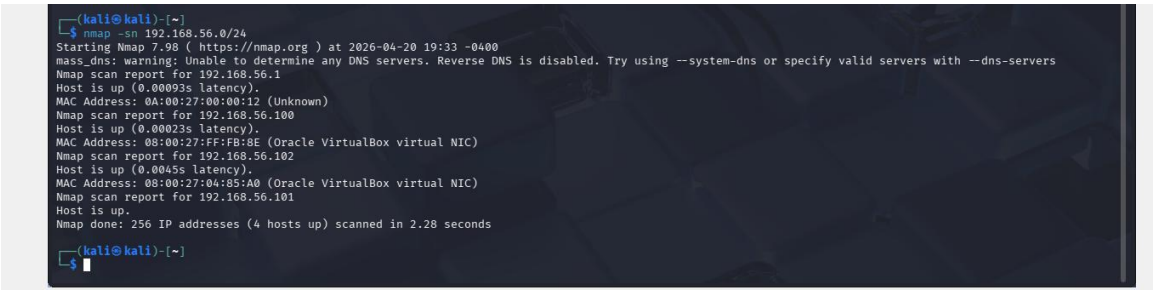

Figure 4: Host Discovery Scan Results (nmap -sn 192.168.56.0/24)

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 19:36 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.00063s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:04:85:A0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 5: Port Scanning Results (nmap 192.168.56.102)

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 19:38 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:04:85:A0 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.08 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 6: Service Enumeration Results (nmap -sV 192.168.56.102)

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O --max-retries 1 --host-timeout 30s 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-20 20:30 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:04:85:A0 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.54 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 7: OS Detection Results (Nmap OS Fingerprinting)

5.2 Traffic Analysis

Wireshark was used to capture ICMP, ARP, and TCP scan traffic. Filters such as 'tcp' were applied to isolate scan behavior. Screenshots added to demonstrate results.

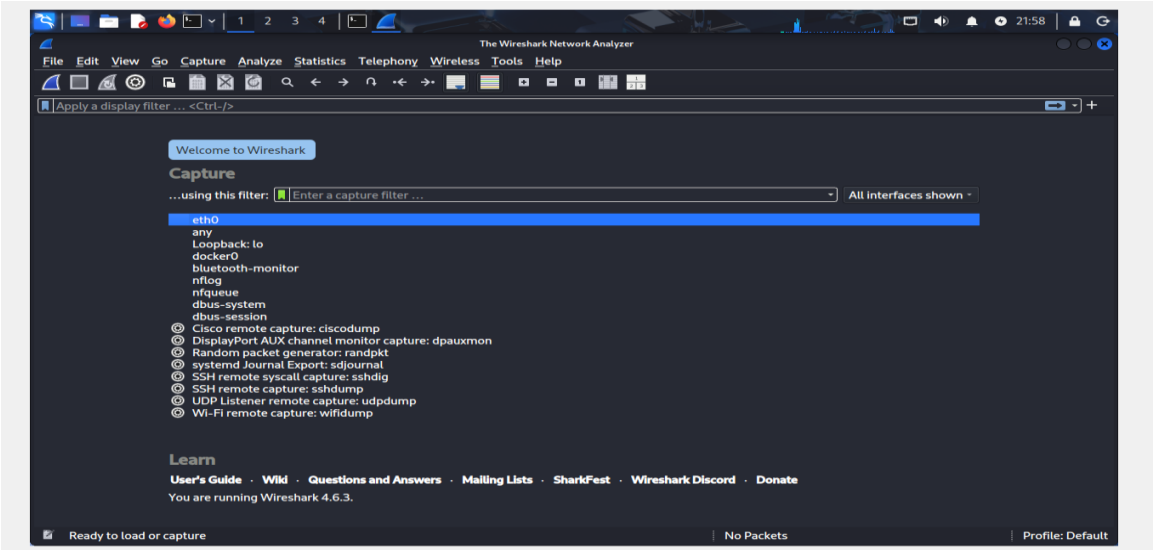

Figure 8: Wireshark Interface Selection Showing eth0

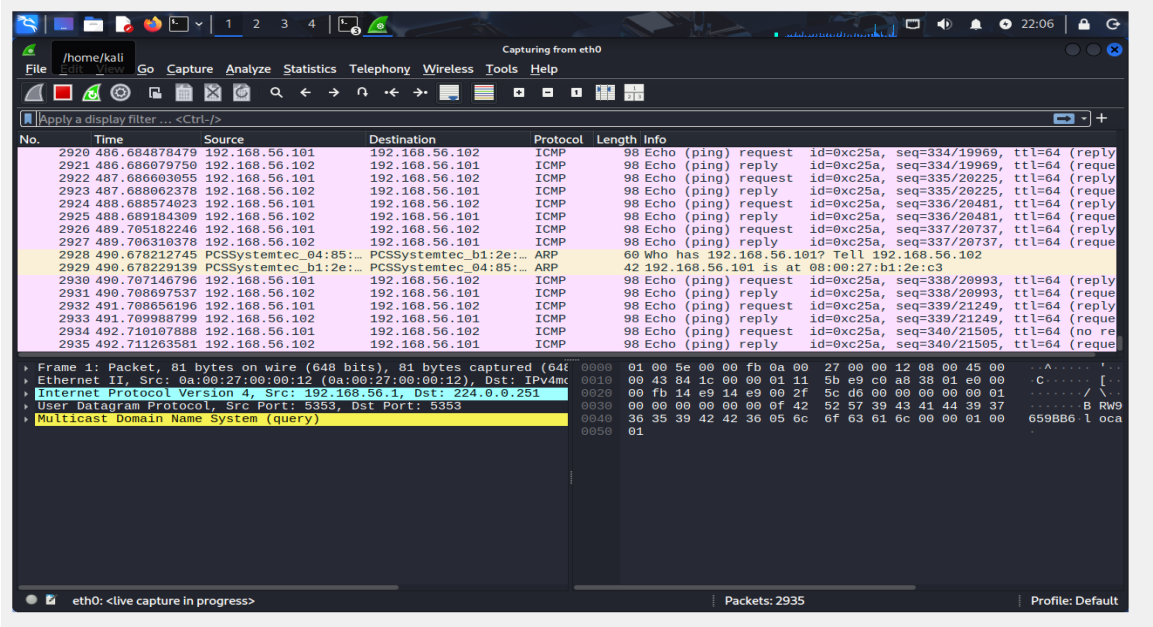

Figure 9: ICMP Echo Request and Reply Packets Between Kali and Metasploitable

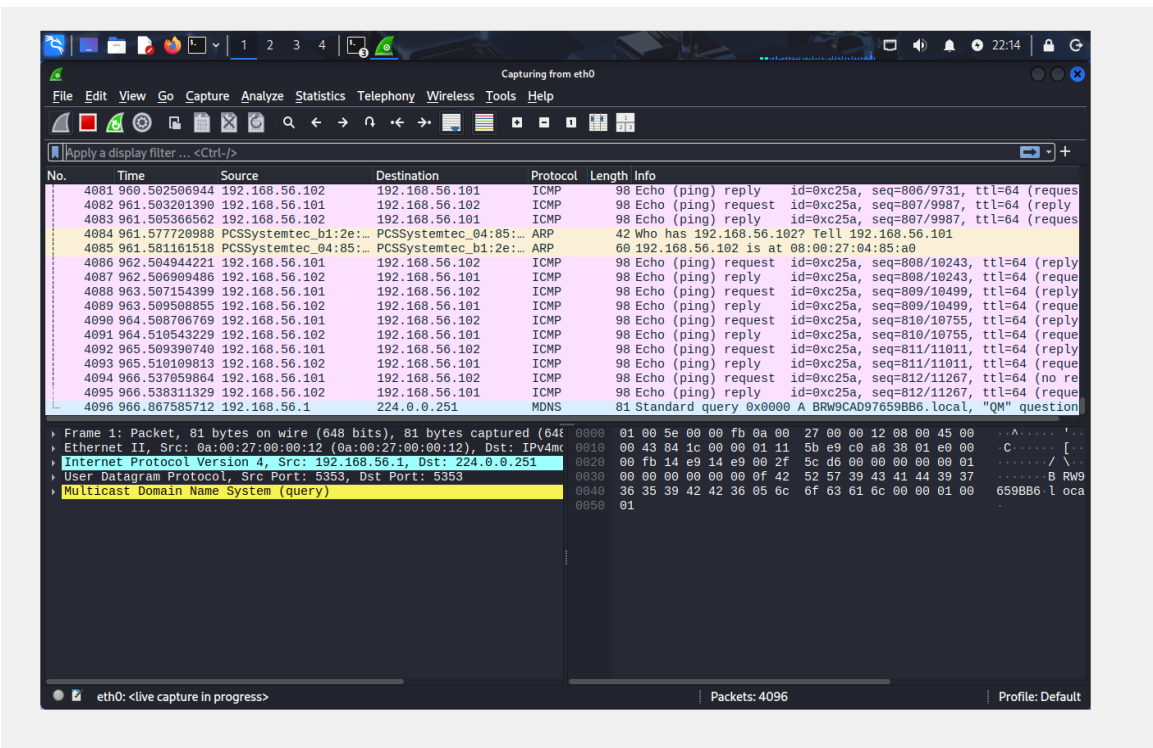

Figure 10: ARP Traffic Showing IP-to-MAC Address Resolution

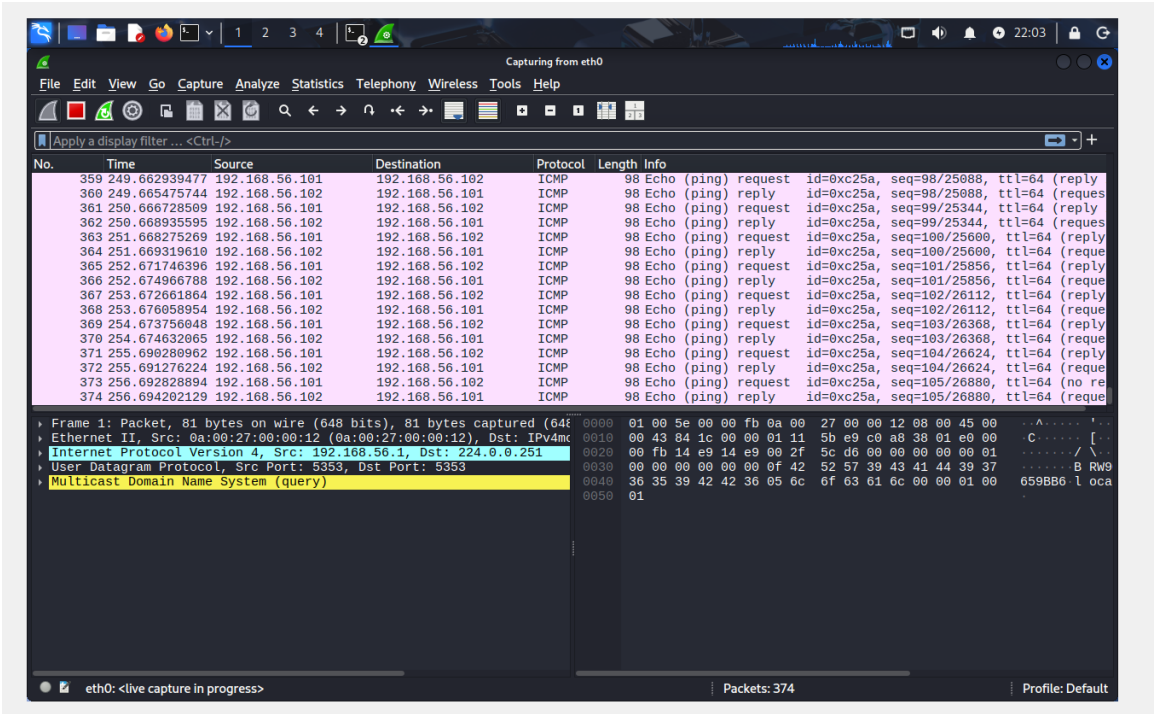

Figure 11: Initial Packet Capture Showing MDNS and Broadcast Traffic

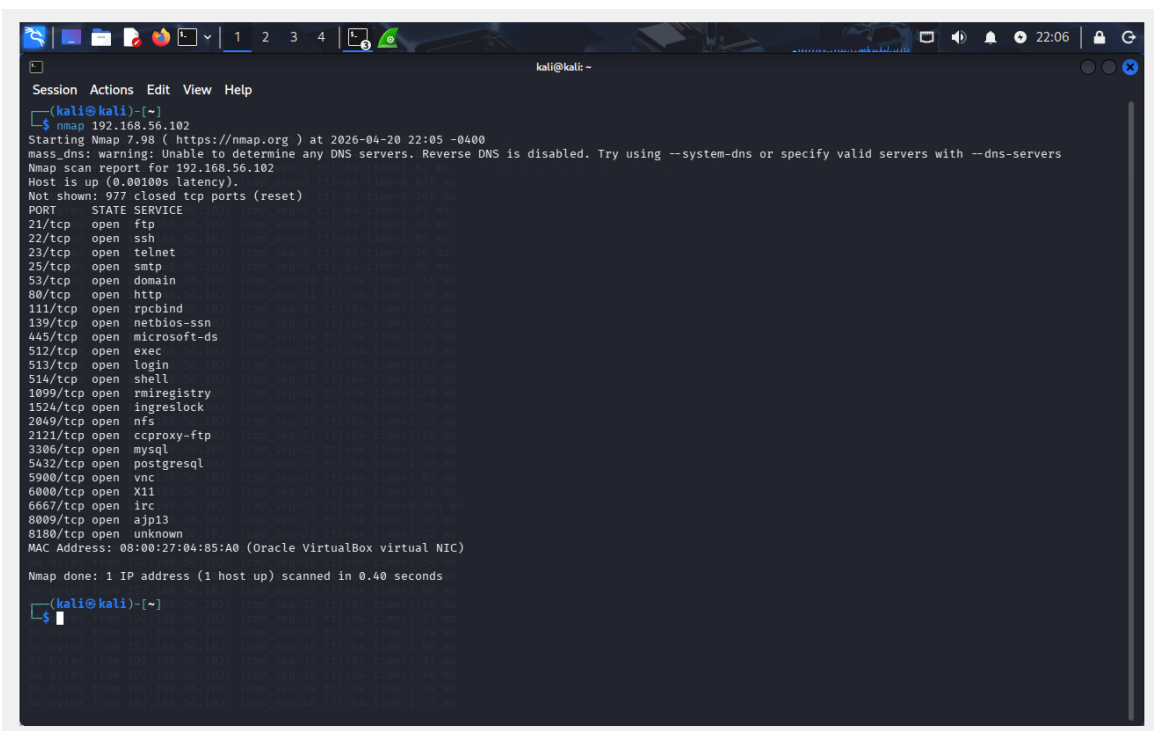

Figure 12: TCP SYN Packets Generated During Nmap Port Scanning

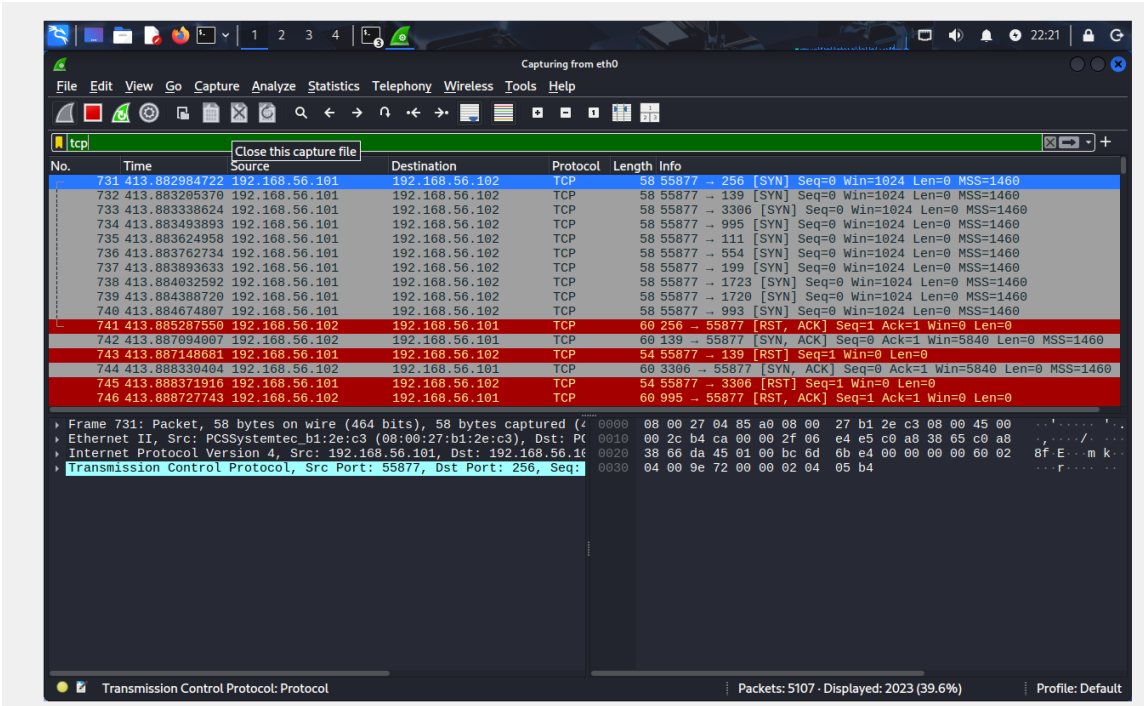

Figure 13: TCP Responses Showing SYN-ACK (Open Ports) and RST (Closed Ports)

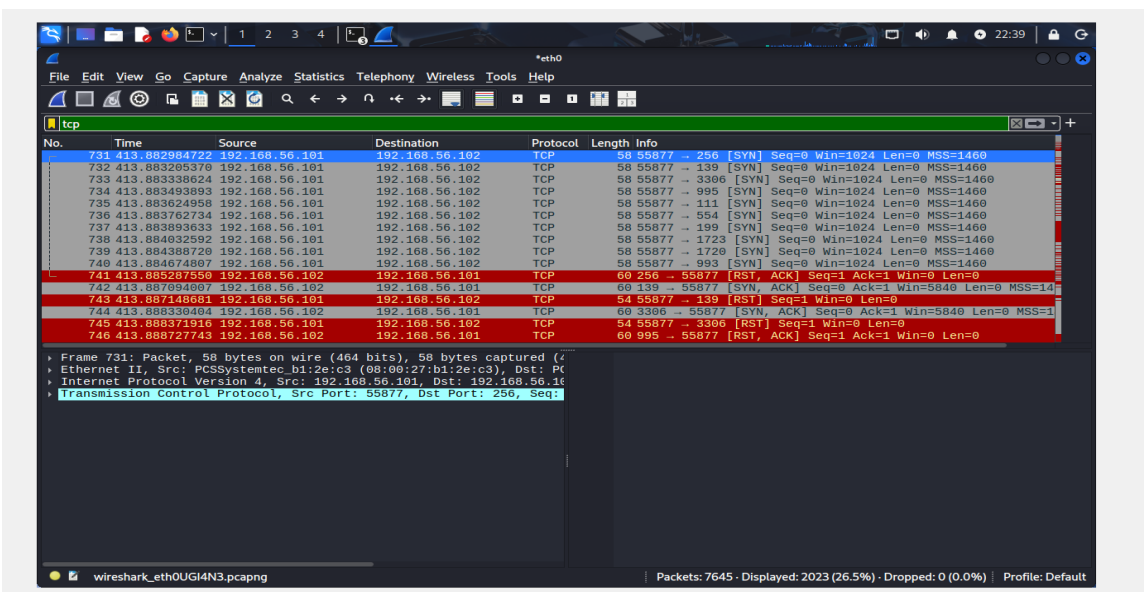

Figure 14: Final Wireshark Capture State with Filtered TCP Scan Traffic.

6. Results and Analysis: The network scanning phase revealed that the target system exposed multiple open ports and services, including FTP (21), SSH (22), Telnet (23), HTTP (80), SMB (139/445), MySQL (3306), and PostgreSQL (5432). Several of these services were outdated or insecure, such as Telnet and older versions of FTP and Apache, indicating a significantly increased attack surface. During traffic analysis, packet captures confirmed scanning behavior through repeated TCP SYN packets sent from the Kali system to multiple target ports. Responses from the target system included SYN-ACK packets for open ports and RST packets for closed ports. This correlation between Nmap results and packet-level evidence demonstrates how reconnaissance activity can be identified both from a tool-based and **a** monitoring perspective.

7. Security Implications: The presence of multiple open and outdated services significantly increases the attack surface of the target system. Services such as Telnet and older SMB implementations are particularly risky due to their lack of encryption and history of vulnerabilities. From a network monitoring perspective, the observed TCP SYN scan behavior strongly indicates reconnaissance activity. A high volume of SYN packets directed at multiple ports within a short time frame is commonly used by attackers to identify accessible services before launching further attacks. Such patterns can be detected using intrusion detection systems (IDS), firewalls, and network monitoring tools, making traffic analysis a critical component of defensive cybersecurity operations

8. Recommendations

• Disable unused services
• Patch outdated systems
• Monitor network traffic
• Implement IDS/IPS
• Restrict access via firewall

9. Conclusion

This project successfully demonstrates the relationship between scanning tools and packet-level evidence, providing a strong foundation for cybersecurity analysis. It highlights the importance of correlating reconnaissance results with network traffic to improve both threat detection and defensive capabilities.